

TAMER ELKOT

AI Engineer

Cairo, Egypt | +20 1097396074 | tamer.elkot.ai@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI Engineer with hands-on expertise in Computer Vision, MLOps, and Backend Development. Experienced in designing and deploying end-to-end AI systems — from data pipelines to production inference servers — using FastAPI, Docker, TensorFlow, and PostgreSQL. Proven ability to lead cross-functional teams and deliver measurable results in real-world environments. Passionate about applying AI to solve impactful problems in agriculture, logistics, healthcare, and security.

TECHNICAL SKILLS

Languages	Python, C, SQL (PostgreSQL)
AI / CV / NLP	TensorFlow, Keras, OpenCV, Transformers, Vision Language Models (VLMs), Scikit-learn, ML Models, Transfer Learning, YOLO, NLP Basics, Prompt Engineering.
Backend	FastAPI, REST APIs, SQLAlchemy, OOP, Streamlit, Django, Alembic. Docker, Linux (Ubuntu), Git, Celery.
Tools	VS Code, Jupyter, Google Colab, Hugging Face, Kaggle, Claude Code, Earth Engine APIs, Roboflow, Postman, Cursor, Zoho API's.

PROFESSIONAL EXPERIENCE

AI & Data Analysis Mentor | Instant Software Solutions *[Part-time] June – Present*

- Mentored a cohort of AI and Data Analysis students across weekly live sessions, translating core machine learning and analytics concepts into hands-on, project-based learning to accelerate job readiness.
- Guided learners through end-to-end data analysis workflows — including Excel, Power BI, and SQL — covering data cleaning, exploratory analysis, dashboarding, and business-insight reporting.
- Reviewed student code and capstone projects, delivering structured feedback on model logic, data pipeline design, and analytical rigor to raise cohort-wide project quality.

AI Engineer Intern | [SEG](#) *[Part-time] Apr 2026 – Present*

- End-to-End Product Ownership – RabiH CRM (رابح): Owned the full product lifecycle from initial requirements gathering with stakeholders through architecture design, data modeling, LLM integration, deployment, and handover.
- Requirements & Architecture: Led discovery sessions to define business requirements, then architected a middleware system bridging Zoho CRM and a React 18 dashboard via FastAPI, PostgreSQL, and WebSocket services to close the gap between raw CRM data and actionable sales decisions.
- Data Modeling & LLM Integration: Designed the underlying data model and integrated a generative AI layer to surface bilingual, context-aware recommendations directly inside the sales workflow, reducing the manual analysis burden on account teams.
- Deployment & Handover: Containerized and deployed the system with Docker and Alembic-managed migrations, then authored technical documentation and led handover sessions to ensure smooth adoption by the operations team.
- Business Impact: Reduced operational overhead and improved sales-pipeline profitability by converting static, manually-reviewed CRM records into a dynamic, predictive decision-support tool for the B2B sales team.

AI Mentor | [Celulla Technology](#) *[Part-time] Apr 2026 – Present*

- **Technical Mentorship & Code Quality:** Lead and mentor a cohort of AI interns by conducting rigorous line-by-line code reviews, ensuring adherence to PEP 8 standards, modular programming, and optimized model performance.
- **Curriculum Development & Technical Support:** Develop and publish specialized academic content covering advanced AI concepts while providing real-time technical troubleshooting to resolve blockers in machine learning pipelines.
- **Project Oversight & Agile Collaboration:** Manage weekly task allocation and progress tracking through sync meetings, facilitating technical discussions to overcome algorithmic challenges and ensure timely project milestones.

Computer Vision Engineer Intern | [Celulla Technology](#) Jan 2026 – Mar 2026

- **Project Lead – Shoplifting Detection:** Directed a team to develop a real-time video classification system using CNN + LSTM (LRCN) and MoViNet architectures to distinguish suspicious from normal behavior.
- **Water Segmentation (Satellite Imagery):** Engineered an AI Water Sentinel tool using Sentinel-2 (12-channel) data and a custom U-Net for global flood monitoring; project was featured on the company's official LinkedIn page.
- **Teeth Classification:** Built an end-to-end dental diagnostic system using FastAPI and Streamlit, supporting real-time switching between a Custom CNN and EfficientNetB0 for multi-class tooth identification from X-rays

Machine Learning Engineer Intern | [Technocolabs Softwares](#) [Remote] Jan 2026 – Present

- **Predictive Risk Engine:** Building an end-to-end SLA breach predictor for logistics.
- **Data Analysis:** Analyzed 200,000+ shipment records; identified Fragility (+14% risk) and Temperature Control (+10% risk) as primary failure drivers.
- **MLOps Pipeline:** Engineered a robust pipeline with data validation and leakage prevention for XGBoost modeling, managed via MLflow experiment tracking and automated CI/CD workflows [verify exact tool/platform].

PROJECTS

[Rabih CRM \(رابع\)](#) | [FastAPI](#) · [PostgreSQL](#) · [Random Forest](#) · [LLMs \(Groq/Llama 3\)](#) · [React 18](#) May 2026

- **Intelligent Middleware Architecture:** Developed an end-to-end AI decision engine bridging Zoho CRM and a React 18 dashboard, transforming static sales pipelines into dynamic, actionable intelligence.
- **Predictive ML Scoring:** Trained and deployed a Random Forest/XGBoost classifier to autonomously predict deal closure probabilities based on historical data, enabling sales teams to prioritize high-value pipelines.
- **Generative AI Agents:** Engineered a dual-LLM pipeline utilizing Groq (Llama 3.3 70B) for structured decision-making and Grok (xAI) for conversational output, utilizing few-shot prompting to generate tailored, bilingual (Arabic/English) "Next Best Action" insights.
- **Automated Workflow Pipeline:** Designed an idempotent PostgreSQL data ingestion pipeline and an APScheduler-driven loop that automatically dispatches customized WhatsApp follow-up alerts for urgent, high-risk deals without manual intervention.

AgriVision | [U-Net](#) · [Sentinel-2 12-Channel](#) · [FastAPI](#) · [Docker](#) · [PostgreSQL](#) Mar 2026

- **Geospatial Pipeline:** Engineered a robust 12-channel multispectral data pipeline using Sentinel-2 imagery with advanced radiometric normalization and augmentation to isolate water signatures across diverse global terrains.
- **Deep Learning Architecture:** Trained a custom U-Net for semantic segmentation with a hybrid Dice + Binary Cross-Entropy loss to mitigate class imbalance and achieve high-precision pixel-level accuracy.
- **MLOps Integration:** Deployed via FastAPI for asynchronous processing and Docker for containerization, ensuring seamless model inference and pipeline portability.
- **Temporal Analytics:** Built an interactive geospatial dashboard for real-time ROI selection and historical tracking, persisting time-series water data in PostgreSQL for agricultural and flood analysis.

EmpVision | [YOLOv8](#) · [FaceNet](#) · [FastAPI](#) · [Docker](#) · [PostgreSQL](#) Feb 2026

- **Biometric System:** Replaced legacy HOG methods with YOLOv8 + FaceNet, achieving high accuracy at 90° angles and in low-light conditions.

- **Backend Architecture:** Designed a FastAPI + PostgreSQL backend using native ARRAY types to store 512-dimensional face embeddings for vector similarity search.
- **Real-Time Streaming:** Designed a multi-threaded RTSP pipeline for zero-latency streaming with Passive Video Enrollment to filter blurred frames.

EDUCATION

Bachelor of Science in Mechatronics Engineering | Higher Institute of Engineering & Technology | *Expected Graduation: May 2027*

CERTIFICATIONS

- [HCIA-AI V4, Huawei ICT Academy](#) – Jul 2025
- [Machine Learning & Deep Learning, NTI](#) – Aug 2025
- [Deep Learning for Computer Vision](#), NTI – Sep 2025
- [Artificial Intelligence, NTI](#) – Jul 2025
- [Machine Learning & Deep Learning, Neurotech](#) – Apr 2025
- Prompt Engineering, Coursera 2026
- [Data Base Design, Coursera](#)
- [Claude Code 101, Anthropic](#)
- [AI internship Certification](#), SEG
- [Excel Certification, DataCamp](#)